

9.3.2 Details of Reset Flag Registers

9.3.2.1 RESF — Reset Factor Register

This register contains information about which type of resets occurred after the last power-on clear reset. This register is initialized by a power-up reset PURES.

Each reset condition sets the corresponding flag in the register.

For example, if a clock monitor reset $\overline{\text{CLMA0RES}}$ occurs after a watchdog timer reset WDTA0RES , RESF reads $0000\ 000\text{A}_{\text{H}}$.

Access: This register is a read-only register that can be read in 32-bit units.

Address: $\text{FFF8}\ 0760_{\text{H}}$

Value after reset: $0000\ 0200_{\text{H}} / 0000\ 0300_{\text{H}}$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	RESF ₁₀	RESF ₉	RESF ₈	RESF ₇	RESF ₆	RESF ₅	RESF ₄	RESF ₃	RESF ₂	RESF ₁	RESF ₀
Value after reset	0	0	0	0	0	0	1	1/0 ^{*1}	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Note 1. For details, see **Figure 9.4, When $\overline{\text{RESET}}$ is Released before Execution of Flash Sequence.**

Table 9.5 RESF Register Contents (1/2)

Bit Position	Bit Name	Function
31 to 11	Reserved	When read, the value after reset is returned.
10	RESF10	Reset flag by DeepSTOP mode 0: No reset occurred 1: Reset has occurred
9	RESF9	Power-up reset flag 0: No reset occurred 1: Reset has occurred
8	RESF8	External reset flag 0: No reset occurred 1: Reset has occurred
7	RESF7	CVM reset flag 0: No reset occurred 1: Reset has occurred
6	RESF6	LVI reset flag 0: No reset occurred 1: Reset has occurred
5	RESF5	CLMA2 reset flag 0: No reset occurred 1: Reset has occurred
4	RESF4	CLMA1 reset flag 0: No reset occurred 1: Reset has occurred
3	RESF3	CLMA0 reset flag 0: No reset occurred 1: Reset has occurred

Table 9.5 RESF Register Contents (2/2)

Bit Position	Bit Name	Function
2	RESF2	WDTA1 reset flag 0: No reset occurred 1: Reset has occurred
1	RESF1	WDTA0 reset flag 0: No reset occurred 1: Reset has occurred
0	RESF0	Software reset flag 0: No reset occurred 1: Reset has occurred